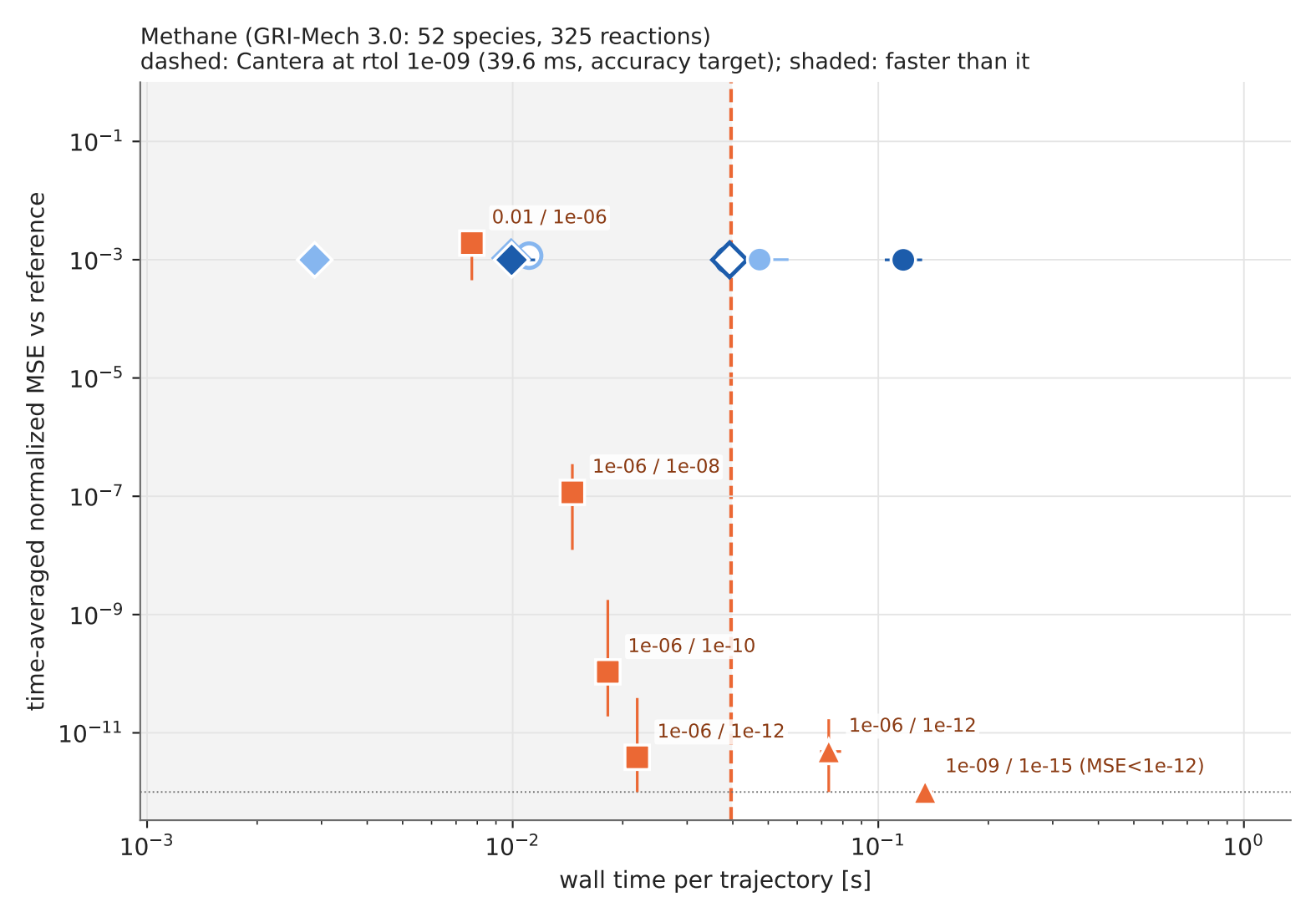

Markers: median over conditions x 25 repetitions; bars: IQR. MSE against the reference at its generation tolerance; values below 1e-12 drawn on the dotted floor.

ChemKAN, PyTorch
ChemKAN, Julia port
dark blue: ChemKAN at its training tolerance
light blue: at the other tolerance (H2/CH4 rtol 1e-2, biodiesel 1e-10)
hollow: all conditions in one batched solve
reference, Python (Cantera / LSODA), labels rtol / atol
reference, Julia (Arrhenius.jl / released kinetics)
Python reference at its generation tolerance